Dataset Generation and Evaluation Platform for Dynamic Graph Neural Network ETA Prediction

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Abstract. This paper presents an integrated platform for developing and evaluating Dynamic Graph Neural Network (GNN) models for Estimated Time of Arrival (ETA) prediction. We describe a vehicle-level dataset that combines synthetic data from microscopic traffic simulation with real-world GPS trajectory data, modeling individual vehicles rather than regional aggregates. The dataset generation pipeline creates route-aware dynamic spatio-temporal graphs, with multi-variant extraction for fair model comparison; a separate conversion pipeline transforms trajectory-based datasets (e.g., Porto taxi) into the dynamic graph format; and an online validation environment enables real-time model testing. We briefly overview the DSTRA-GNN architecture that consumes the generated datasets. The system addresses critical challenges in both dataset quality and model evaluation for intelligent transportation research.

Keywords: Dataset Generation · Graph Neural Networks · ETA Prediction · Vehicle-Level Modeling · Online Model Validation · Route-Aware Graphs

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